

Problem 2

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Problem 1 Differentiate the function f defined by $f(x) = 1/x^8$ in two different ways.

Explanation. Applying the quotient rule:

$$\begin{aligned} f'(x) &= \frac{(1)' \cdot x^8 - (1 \cdot (x^8)')}{(x^8)^2} \\ &= \frac{(0 \cdot x^8) - (1 \cdot 8x^7)}{x^{16}} \\ &= \frac{-8x^7}{x^{16}} \\ &= \frac{-8}{x^9}. \end{aligned}$$

Applying the power rule:

$$\begin{aligned} f(x) &= \frac{1}{x^8} = x^{-8} \\ \implies f'(x) &= (x^{-8})' \\ &= -8 \cdot x^{-8-1} \\ &= -8x^{-9} \end{aligned}$$